OBSERVATION

Number of turns of the coil n =

Circumference of the coil (2πr) =

Radius of the coil r =

S.No	Current I (A)	Deflection in TG (degree)				Mean θ (degree)	$k=\frac{I}{\tan\theta}$
		θ_1	θ_2	θ_3	θ_4		
1							
2							
3							
4							
						Mean	

CALCULATION

$$B_H = \frac{\mu_0 n k}{2r} =$$

RESULT

The horizontal component of Earth's magnetic field is found to be _____

Note:

i) The magnetic materials and magnets present in the vicinity of TG should be removed.

ii) The readings from the ends of the aluminium pointer should be taken without parallax error.

iii) The deflections of TG is restricted between 30° and 60°. It is because, the TG is most sensi tive for deflection around 45° and is least sensitive around 0° and 90°. We know that

$$I = k \tan\theta$$

$$\text{or } dI = k \sec^2\theta \, d\theta$$

$$\frac{d\theta}{dI} = \frac{\sin 2\theta}{2I}$$

For given current, sensitivity $\frac{d\theta}{dI}$ is maximum for $\sin 2\theta = 1$ or $\theta = 45°$

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